

Key Pillar Ai

Audit Log & Task Tracking App

App Structure and Important Requirements

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1. App Overview

This app is a simple internal project-based task tracking system for Key Pillar Ai. It works like a controlled to-do list where the admin creates and assigns tasks to developers, developers update their own assigned tasks, and the system keeps permanent history and audit logs for admin review.

Main Purpose

Admin controls projects and task assignment. Developers work on their assigned tasks, update status, and comment under tasks when they face issues. Every important action is saved in history and audit logs.

2. Major Parts of the App

- Authentication - password-only login controlled by admin.
- Admin Area - project, task, user, password, history, and audit log management.
- Developer Area - assigned task list, task editing, status updates, and comments.
- Project Module - project details, task list, and project progress.
- Task Module - task details, assigned developer, status flow, and comments.
- Task History Module - permanent record of task changes visible only to admin.
- Audit Log Module - automatic tracking of important system actions.
- QR Install/Access Module - QR code opens the app access/install page.

3. User Roles

Role	Main Responsibility	Access Level
Admin	Controls users, projects, tasks, assignments, passwords, history, and audit logs.	Full access
Developer	Works on assigned tasks, edits own task details, updates status, and adds comments.	Limited access

4. Authentication Structure

The app uses a simple password-only login model. Users log in using their account password. Developer account passwords are managed by the admin.

- Users can log in with password only.
- Admin has access to developer account passwords and login control.
- Developers cannot change their own passwords.
- Only the admin can change developer passwords.
- Admin can create, update, and control developer login accounts.

5. Admin Area

The admin is the main controller of the system. The admin controls task creation, task assignment, user accounts, password changes, task history, and audit logs.

Admin Functionality	Description
Manage users	Create developer accounts, update accounts, deactivate users, and manage passwords.
Manage projects	Create, edit, archive, or delete projects.
Create tasks	Only admin can create tasks.
Assign tasks	Only admin can assign tasks to developers.
Edit tasks	Admin can edit any task.
Delete tasks	Admin can delete any task. Deleted records remain in history.
Approve tasks	Admin approves completed work and changes status to Complete.
View history	Admin can view full task history.
View audit logs	Admin can view all audit logs.

6. Developer Area

Developers have access only to their assigned work. They can edit their own assigned tasks, update task progress, mark tasks as failed or waiting for approval, and add comments under tasks.

Developer Functionality	Description
View assigned tasks	Developer can see only tasks assigned to them.
Edit own tasks	Developer can edit their own assigned task details.
Update status	Developer can move task status from Pending to In Progress, Failed, or Waiting for Approval.
Add comments	Developer can comment below their own tasks.
Report issues	Developer can explain the issue they faced by commenting under the task.
Delete own tasks	Not allowed. Developers cannot delete their own tasks.
Manage assignments	Not allowed. Developers cannot assign or reassign tasks.

7. Task Status Flow

The app has a 5-step task progression flow. This flow shows the full journey of a task from assignment to admin approval.

Step	Status	Who Sets It	Meaning
1	Pending	Admin/System	The admin has created and assigned the task to a developer.
2	In Progress	Developer	The developer has started working on the task.
3	Failed	Developer	The developer cannot complete the task and must explain the issue in comments.
4	Waiting for Approval	Developer	The developer has completed the work and is waiting for admin review.
5	Complete	Admin	The admin has reviewed and approved the task.

8. Task Status Rules

- Newly assigned tasks start with the Pending status.
- When the developer starts the task, the status becomes In Progress.
- If the developer cannot complete the task, the status can be changed to Failed.
- When the developer finishes the task, the status becomes Waiting for Approval.
- Only the admin can approve the task and change the status to Complete.
- If a task is marked Failed, the developer should add a comment explaining the issue.
- A project can be considered complete only when its active tasks are marked Complete.

Important Status Logic

Developers do not directly mark a task as Complete. They mark it as Waiting for Approval. The task becomes Complete only after admin approval.

9. Project Structure

A project is the main container for tasks. Each project contains a list of tasks assigned by the admin to developers.

Project Field	Description
Project name	Name of the project.
Description	Short details about the project.
Project status	Not Started, In Progress, Completed, or Archived.
Created by	Admin who created the project.
Created date	Date and time the project was created.
Tasks	List of tasks under the project.
Progress	Based on task completion and admin approval.

10. Task Structure

A task is a single work item inside a project. Each task is assigned by the admin to one developer.

Task Field	Description
Task title	Name of the task.
Task description	Details about what the developer needs to do.
Project name	The project where the task belongs.
Assigned developer	Developer responsible for the task.
Status	Pending, In Progress, Failed, Waiting for Approval, or Complete.
Due date	Expected completion date, if provided.
Comments	Regular updates or issue explanations.
Created by	Admin who created the task.
History	Permanent record of changes.

11. Task Comments and Issue Notes

Each task has a comment section below it. Developers use comments to explain progress, problems, or reasons for failure.

- Developers can add comments under their own assigned tasks.
- Admin can view comments under all tasks.
- Comments show the writer, date, time, and message.
- When a task is marked Failed, the developer should explain the issue in a comment.
- Issue comments help the admin understand why the task could not be completed.

Comment Type	Purpose	Example
Normal comment	Progress update or regular note.	I have completed the UI part and started API integration.
Issue comment	Explains a problem or blocker.	The API is returning an error, so I cannot finish the task.
Failure explanation	Explains why task was marked Failed.	I could not complete this because the required credentials were not provided.

12. Task History

Task history is a permanent record of changes made to a task. It cannot be deleted by developers and can only be viewed by the admin.

- Task created
- Task assigned
- Task edited
- Task status changed
- Comment added
- Comment edited
- Task marked Failed
- Task marked Waiting for Approval
- Task approved as Complete
- Task deleted by admin

13. Audit Logs

Audit logs are automatic system records of important actions. They are separate from normal comments and task history. Only the admin can view audit logs.

Audit Log Event	Tracked Information
User login	User account, role, date, and time.
Password changed	Admin action, affected user, date, and time.
Project created or edited	Project name, admin user, old value, new value.
Task created	Task title, project, assigned developer, created by.
Task assigned	Task title, assigned developer, admin user.
Task edited	Changed fields, old values, new values.
Task status changed	Old status, new status, user, date, and time.
Comment added	Task, comment writer, date, and time.
Task approved	Admin user, task, approval date, final status.

14. Permission Structure

Functionality	Admin	Developer
Login with password	Yes	Yes
Change own password	Yes	No
Change developer password	Yes	No
View all developer accounts	Yes	No
Create project	Yes	No
Edit project	Yes	No
Delete/archive project	Yes	No
Create task	Yes	No
Assign task	Yes	No
Reassign task	Yes	No
View all tasks	Yes	No
View own assigned tasks	Yes	Yes
Edit any task	Yes	No
Edit own assigned task	Yes	Yes
Delete own task	No need	No
Delete any task	Yes	No
Add comment to own task	Yes	Yes
Mark task as Failed	Yes	Yes, own task only
Mark task as Waiting for Approval	Yes	Yes, own task only
Approve task as Complete	Yes	No
View task history	Yes	No
View audit logs	Yes	No

15. Admin Dashboard

The admin dashboard gives the admin a full overview of projects, tasks, developers, task statuses, comments, history, and audit logs.

- Total projects
- Active projects
- Completed projects
- Total tasks
- Pending tasks
- In Progress tasks
- Failed tasks
- Waiting for Approval tasks
- Complete tasks
- Developer list
- Recent comments
- Recent task activity
- Recent audit logs

16. Developer Dashboard

The developer dashboard shows only the tasks assigned to the logged-in developer.

- My assigned tasks
- Pending tasks
- In Progress tasks
- Failed tasks
- Waiting for Approval tasks
- Complete tasks
- Task comments
- Issue comments
- Task due dates
- Project names

17. Main App Pages

Page	Purpose
Login page	Users log in with password.
Admin dashboard	Full system overview for admin.
Developer dashboard	Developer-specific task overview.
Projects page	Project list for admin.
Project details page	Project information and tasks.
Task details page	Task information, status, and comments.
My Tasks page	Developer view of assigned tasks.
Task history page	Admin-only history view.
Audit logs page	Admin-only system audit trail.
Users page	Admin user and password management.
QR access page	QR code for app access/install page.

18. QR Code Access Structure

The app will include a QR code that opens the app access or install page. This allows developers to scan the QR code and open the app quickly on their device.

- QR code opens the app link or install/access page.
- User scans QR code from mobile device.
- User opens login page.
- User logs in with password.
- Developer sees assigned tasks after login.
- Admin sees admin dashboard after login.

19. Core Data Structure

The app will mainly store users, projects, tasks, comments, task history, and audit logs.

Data Type	Key Fields
Users	User ID, name, password, role, status, created date, last login.
Projects	Project ID, project name, description, status, created by, created date.
Tasks	Task ID, project ID, title, description, assigned developer, status, due date, created by, updated date.
Task comments	Comment ID, task ID, user ID, comment text, comment type, created date.
Task history	History ID, task ID, changed by, action type, old value, new value, date and time.
Audit logs	Audit ID, user ID, action, entity type, entity ID, old value, new value, date and time.

20. Main App Logic

- Only admin can create projects.
- Only admin can create tasks.
- Only admin can assign tasks to developers.
- Developers can only see their own assigned tasks.
- Developers can edit their own assigned tasks.
- Developers cannot delete their own tasks.
- Developers can add comments and issue notes under their own tasks.
- Developers can mark tasks as In Progress, Failed, or Waiting for Approval.
- Only admin can approve tasks as Complete.
- Only admin can delete tasks.
- Only admin can view task history.
- Only admin can view audit logs.
- Every important action creates audit logs automatically.

21. Full App Structure

The app is organized into clear modules. Each module has a specific responsibility inside the system.

Main Area	Included Parts
Authentication	Password-only login, admin-managed passwords, no developer password change.
Admin Area	Admin dashboard, users, passwords, projects, tasks, assignments, approvals, history, and audit logs.
Developer Area	Developer dashboard, My Tasks, edit own tasks, status updates, comments, and issue notes.
Project Module	Project list, project details, project tasks, project progress, and project status.
Task Module	Task details, assigned developer, five-step status flow, comments, and issue notes.
History and Audit Module	Task change history, status history, comment history, and system audit logs.

22. Major Highlights

- The app is a simple internal to-do list system with audit logs.
- Admin has full control over users, passwords, projects, tasks, assignments, approvals, history, and audit logs.
- Developers log in with password only and cannot change their own passwords.
- Only admin can change developer passwords.
- Only admin can create and assign tasks.
- Developers can edit their own assigned tasks but cannot delete tasks.
- Task progression has five statuses: Pending, In Progress, Failed, Waiting for Approval, and Complete.
- Developers can mark tasks as Failed when they cannot complete the task.
- Developers mark finished work as Waiting for Approval.
- Only admin can approve the task and mark it Complete.
- Developers can comment below tasks to explain issues or progress.
- Only admin can see full task history and audit logs.
- The QR code gives fast access to the app from a device.

Final App Summary

Key Pillar Ai Audit Log & Task Tracking App is a controlled internal task management system where admin creates and assigns work, developers update their assigned tasks, issues are captured through comments, and all major activity is saved in admin-only history and audit logs.